Predicting Floc Size in South San Francisco Bay from Waves and Currents

ML Mini Project write-up. A physics-only replication of Egan et al. (2022, JGR Oceans) and Egan's CS229 report, built on the public Stanford Digital Repository dataset.

1. Problem statement

Mud in estuaries travels as **flocs**: loose clumps of fine particles and organic matter. Floc size sets how fast mud sinks (settling speed grows with diameter squared), so it controls where sediment and attached pollutants end up. Sediment models need it, but it is rarely measured and varies about threefold within a season. **We ask: can floc size be predicted from wave and current measurements alone, and does the prediction hold on days and seasons the model has not seen?** This is a supervised regression task: tabular physical inputs, one continuous output (mean floc diameter in micrometres).

2. Dataset

NSF OCE-1736668 field campaign (Stanford Digital Repository, purl wv787xr0534): three deployments of about four weeks (Summer 2018, Winter 2019, Spring 2019) on 2 m deep shoals. From the 2.5 GB archive we extracted only what was needed, using HTTP range requests.

Instrument	Measures	Rate	Used for
LISST laser sizer (P2, 35 cm above bed)	Particle size distribution, 32 bins	60 s	Target d
ADV current meter (P2, 15 cm)	3D velocity, pressure	8 Hz, 14 min/h	Waves, tide, turbulence
CTD sensors (P2, P1)	Temperature, salinity	60 s	T, S

After quality control: **1,120 hourly samples** (Summer 427, Winter 406, Spring 287) from 1,666 bursts. Not available in this archive: the chlorophyll and light-absorption (biology) sensors both papers used. Data issues we found and documented: the ADV pressure is not in the units the README states, P2 salinity is unusable (P1 used instead), and low-quality ADV samples are NaN rather than interpolated.

3. Approach

- **Target.** Average the 1-minute laser readings over each 14-minute ADV burst, drop saturated optics (transmission below 0.1), remove isolated edge-bin spikes (the paper's 40% rule) and take the volume-weighted mean diameter. Resulting medians (83, 62, 106 micrometres) match the paper's Fig. 1.
- **Features (6).** Wave orbital velocity u_b (from the velocity spectrum, cross-checked against linear wave theory, r = 0.95), tidal current now and 4 hours earlier, turbulence dissipation, temperature, salinity. Gaps are never filled; bad bursts are dropped.
- **Model.** A **Random Forest**, the model of the internal CS229 study, so results compare directly. It suits small, noisy, non-linear tabular data with mixed units, needs no scaling and reports feature importance. LASSO with recursive feature elimination (journal paper) and SVR (CS229) are run for comparison.
- **Evaluation.** Hourly floc size is strongly autocorrelated, so a random split lets the model see the neighbours of every test hour. We therefore report **time-blocked** cross-validation (contiguous multi-day blocks, 12 h gap) and **leave-one-season-out**, alongside each paper's own protocol. Metric: r^2 (1 = perfect, 0 = no better than the mean) with day-level bootstrap intervals and RMSE.

4. Implementation overview

Python 3.12 with numpy, scipy, pandas, scikit-learn and matplotlib, in a project virtual environment. Raw MATLAB files are read with scipy.io.loadmat; the feature table is cached as Parquet so models run in seconds. Random seeds are fixed, so every number in this write-up is reproducible.

File	Role
build_features.py	Builds the one-row-per-burst feature table from the raw files (about 30 s)
run_random_forest.py	Main demo: trains the Random Forest and prints our scores next to both papers
src/load.py, features.py	Loading, time conversion, QC, spectra, despiking, target and features
src/models.py, predict.py	Time-blocked splits, LASSO/RF/SVR, scoring; single-row prediction with range checks
scripts/	All-model runs, figures, diagnostics; outputs/ holds tables, figures and the full report

Live demonstration (two commands):

```
ven/bin/python build_features.py
ven/bin/python run_random_forest.py
```

5. Results

Setup	r^2 (ours)	Reference
Class-paper setup: all seasons, random 70/30, RF 32 trees, d50	0.60	CS229 reported 0.81
Class-paper SVR (C = 2048, epsilon = 4), d50	0.53	CS229 reported 0.76
Our 6 features, all seasons, random split	0.64	optimistic protocol
Our 6 features, all seasons, held-out time blocks	**0.34** (95% CI 0.15 to 0.49)	honest estimate
One season at a time, held-out time blocks	below 0 (best -0.14)	no skill within a season
Predict an unseen season (leave-one-season-out)	-0.1 to -4.6	fails on new conditions
LASSO fitted on its own data (journal protocol)	0.17 to 0.29 per season, 0.43 pooled	journal 0.15 to 0.65

What drives floc size. Permutation importance ranks the tidal current 4 hours earlier first (positive effect) and wave velocity second (negative effect), matching the journal paper. Turbulence adds almost nothing and salinity is weak within a season, contrary to the CS229 report, which ranked salinity first. The model follows the tide-to-tide rhythm of floc size but misses multi-day shifts in the average.

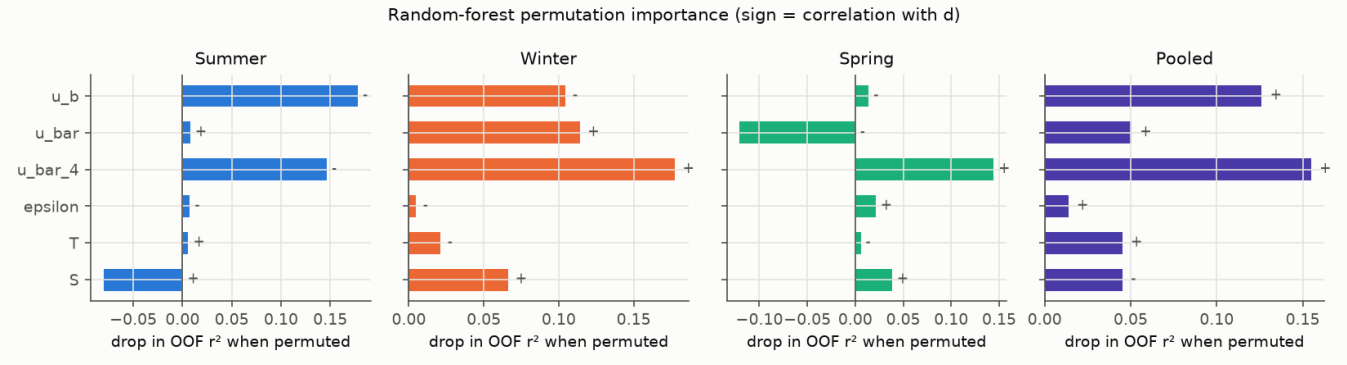


Figure: Random Forest permutation importance per season (drop in held-out r^2 when a feature is shuffled; + / - shows the direction of its effect on floc size).

6. Conclusions

- The physical relationships reported in the journal paper reproduce on independent public data: waves break flocs and the delayed tide brings larger ones.
- Physics alone explains about a third of floc-size variation across seasons (r^2 = 0.34) and little within a season. Measured the way each paper measured it, our scores sit close to theirs (0.60 vs 0.81; 0.17 to 0.43 vs 0.15 to 0.65).
- The remaining gap has three causes: missing biology inputs (particle type, chlorophyll), the scoring method (random splits and in-sample fits reward memorising neighbouring hours) and the measurement location (35 cm at P2 vs 15 cm on the optics platform).
- For a model that lasts years, new data matters more than a new algorithm: biology and sediment sensors, more deployments, and occasional direct size measurements to re-anchor predictions. A next step is the same team's Dryad optics dataset, which adds the particle-type index at the same site.